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Polymerization-induced self-assembly of *N*-substituted glycine *N*-carboxyanhydrides in selected solvents

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ABSTRACT

Poly(sarcosine) (PSar)-based diblock copolypeptoid nanoparticles are promising alternatives to PEGylated polymeric nanocarriers. However, their preparation was limited by the slow polymerization kinetics and low assembly concentrations. Herein, we report the facile preparation of copolypeptoid nanoparticles through polymerization-induced self-assembly of *N*-substituted glycine *N*-carboxyanhydrides (NNCAs) in solvents with low polarity and weak hydrogen-bonding ability like dichloromethane (DCM). In contrast to the conventional polymerization in *N*,*N*-dimethylformamide (DMF), NNCAs were polymerized at an accelerated rate in DCM that minimized side reactions from water and other impurities. With isopentylamine-initiated PSar as the first stabilizing block, the polymerization of *N*-benzyl glycine *N*-carboxyanhydrides resulted in the formation of spherical or rod-like nano-assemblies at high solid contents (up to 15%) within several hours. We believe this work not only improves our basic understanding on NNCA polymerization behaviours, but also offers an efficient strategy to prepare drug nanocarriers with PSar corona, facilitating the downstream application studies.

1. Introduction

The modification of drug nanocarriers with poly(ethylene glycol) (PEG), also well-known as PEGylation, is one of the most successful approaches in drug delivery [1,2]. The incorporation of PEG not only improves the stability of the nanocarriers, but also significantly prolongs their *in vivo* circulation time. Nevertheless, recent studies revealed that PEGylation strategy was limited by the anti-PEG immune response, which reduced the clinical efficacy of PEGylated nanomedicine [3,4]. As a result, several polymers have been studied as alternatives to PEG [5–7]. Among these polymers, poly(sarcosine) (PSar), a hydrophilic polypeptoid, serves as a promising candidate due to its low immunogenicity [8–10]. Nanocarriers based on PSar block copolymers and PSar-lipid hybrids have been extensively studied [11,12], which showed similar drug protections and stealth effects compared to their PEGylated analogues [13–15].

Despite the promising properties of PSar nanocarriers, their biomedical application is limited by the preparation methods. The synthesis of PSar segment is usually achieved through the ring-opening polymerization (ROP) of the corresponding *N*-substituted glycine *N*-carboxyanhydride (NNCA), sarcosine *N*-carboxyanhydride (Sar-NCA) [16–19]. Therefore, the preparation of block copolymers with non-peptoid segments required the design of bifunctional initiators or the modification of polymer chain-end [8]. Meanwhile, while the preparation of amphiphilic block copolypeptoids was successful [15,20], conventional synthetic strategies suffered from relatively long polymerization time, especially for monomers with bulky *N*-substitutions. Even though chain termination was absent during NNCA polymerization [8,21–23], the elongated polymerization process was still vulnerable to various side reactions caused by impurities like water, validating the necessity to develop accelerated polymerization systems [24,25]. Additionally, conventional assembly of PSar-based diblock copolymers was usually prepared in low concentrations. While the recent advances in polymerization-induced self-assembly (PISA) afforded assemblies at high solid contents [26–29], PSar-related PISA was only reported with hybrid polymers [30–32].

Herein we reported the preparation of PSar-based, amphiphilic diblock copolypeptoid nanoparticles through PISA in selected solvents. The use of dichloromethane (DCM) rather than conventional solvents like *N*,*N*-dimethylformamide (DMF) significantly accelerated the polymerization kinetics, leading to well-defined polypeptoids even under open-air conditions. The synthesis of copolypeptoid nano-assemblies at

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high solid contents (up to 15 %) was therefore completed within several hours in one pot. Nano-assemblies with spherical or rod-like morphology were obtained, which exhibited excellent resistance to common proteases for at least 2 weeks, suggesting their potentials as drug nanocarriers for biomedical application studies.

2. Experimental section

2.1. Materials

Chemicals were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd (Shanghai, China) and used as received unless otherwise specified. Organic solvents were obtained from Chinasun Specialty Products Co., Ltd. (Changshu, China). Anhydrous DCM was treated with 4 Å molecular sieves. Chromatographic grade DMF for gel permeation chromatography (GPC) was purchased from Oceanpak (Tianjin, China). Deuterated solvents were purchased from Cambridge Isotope Laboratories, Inc. (Tewksbury, USA). Propylene oxide (PO) was purchased from Energy Chemical (Shanghai, China). *N*-Benzyl-*N*-(*tert*-butoxycarbonyl)glycine and Proteinase K were purchased from Shanghai Macklin Biochemical Technology Co., Ltd. (Shanghai, China). Poly(amido amine) (PAMAM) dendrimer was purchased from MilliporeSigma (St. Louis, USA). Dialysis bag with a molecular-weight cut-off (MWCO) of 3.5 kDa was purchased from Shanghai Yuanye Bio-Technology Co., Ltd (Shanghai, China). Trypsin was purchased from Cytiva (Shanghai, China). Dulbecco's modified eagle medium (DMEM) and fetal bovine serum (FBS) were purchased from Thermo Fisher Scientific Inc. (Waltham, USA).

RAW 264.7 cells (mouse monocyte macrophages) were purchased from the American Type Culture Collection (ATCC). The cells were cultured in DMEM containing 1 % nonessential amino acids, 10 % FBS, 1 % L-glutamine, and penicillin–streptomycin cocktail.

2.2. Instruments

Nuclear magnetic resonance (NMR) spectra were recorded on a Bruker Ascend 400 MHz spectrometer at Soochow University. The chemical shifts (δ) were reported in ppm and referenced to the residual protons in the deuterated solvents. MestReNova software (version 14.0.0, Mestrelab Research, Escondido, USA) was used for all NMR analysis. Fourier transform infrared (FTIR) spectra were recorded using a Thermo Fisher Nicolet iS20 FTIR spectrometer. GPC experiments were performed on a system equipped with an isocratic pump (1260 Infinity II, Agilent, Santa Clara, USA), a multi-angle static light scattering (MALS) detector (DAWN, Wyatt Technology, Santa Barbara, USA), and a differential refractive index (dRI) detector (Optilab, Wyatt Technology, Santa Barbara, USA). The detection wavelength of the MALS detector was set at 658 nm. Separations were performed using serially connected size exclusion columns (KD-803, KD-804, and KD-806, 8 × 300 mm, Shodex, Yokohama, Japan) using DMF containing 0.1 M LiBr as the mobile phase at a flow rate of 1 mL/min at 60 °C. The MALS detector was calibrated using pure toluene and can be used for the determination of the absolute molecular weights (MWs). The MWs of polymers were determined based on the d*n*/d*c* value of each polymer sample calculated offline by using the internal calibration system processed by the ASTRA 8 software (version 8.1.0, Wyatt Technology, Santa Barbara, USA). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) was analyzed on a Bruker Rapiflex with *trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB) as the matrix. Dynamic light scattering (DLS) experiments were performed using a Zeta-sizer Nano ZS90 (Malvern Panalytical Ltd, Malvern, UK) with a He-Ne laser (λ = 633 nm) at a scattering angle of 90° at 25 °C. Transmission electron microscopy (TEM) images were collected on a FEI Tecnai 20 transmission electron microscope (FEI Nano Ports, Hillsboro, USA) with an acceleration voltage of 200 kV. The samples were prepared by dropping the polymer solutions (10 μL, 1.0 mg/mL) onto the copper grids and drying at room temperature overnight. Scanning electron microscopy (SEM) images were recorded with ZEISS Gemini 500 scanning electron microscope (Carl Zeiss, Oberkochen, Germany). Circular dichroism (CD) data was collected using a chirascan spectrometer V100 (Applied Photophysics, Leatherhead, UK). The pathlength of the CD cuvette was 1 mm.

2.3. Synthesis of Sar-NCA

Sar-NCA was prepared according to literature procedures [30]. The grounded sarcosine (3.0 g, 0.034 mol) was added into a flask and dried under vacuum for 2 h, into which tetrahydrofuran (THF) (100 mL) and α-pinene (7.0 g, 0.051 mol) were added. Triphosgene (4.0 g, 0.014 mol) was then added to start the reaction, which was stirred at 50 °C for 3 h. The solvent was removed under vacuum, resulting in brown oil as the crude product. After one-time recrystallization in DCM/*n*-hexane (1:4, v/v), Sar-NCA was obtained as transparent crystals after sublimation at 70 °C under vacuum (Yield: 60 %). ^{1}H NMR (400 MHz, $CDCl_3$, δ): 4.13 (s, 2H, ring CH_2), 3.05 (s, 3H, CH_3). FTIR (neat, υ, cm^{-1}): 1853, 1786 (C = O).

2.4. Synthesis of N-benzyl glycine N-carboxyanhydride (N-Bz-NCA)

N-Bz-NCA was synthesized in the presence of PO as acid scavengers [33]. *N*-Benzyl-*N*-(*tert*-butoxycarbonyl)glycine (1.0 g, 3.77 mmol) was dissolved in THF (50 mL), into which PO (2.64 ml, 37.7 mmol) and triphosgene (534 mg, 1.89 mmol) were added. The reaction was stirred at room temperature for ~ 24 h. After the solvent removal under vacuum, the resulting light-yellow solid was recrystallized three times in THF/*n*-hexane (1:20, v/v) to obtain a white needle-like solid (Yield: 70 %). ^{1}H NMR (400 MHz, $CDCl_3$, δ): 7.50–7.20 (m, 5H, ArH), 4.56 (s, 2H, ArCH_2-), 3.95 (s, 2H, ring CH_2). ^{13}C NMR (100 MHz, $CDCl_3$, δ): 165.3, 152.3, 133.9, 129.5, 129.0, 128.5, 48.4, 47.8. FTIR (neat, υ, cm^{-1}): 1853, 1784 (C = O).

2.5. Polymerization kinetics

The polymerization kinetics was monitored *in situ* in DCM or DMF by FTIR. Taking *n*-hexylamine (Hex-NH_2)-initiated polymerization in DCM as an example, Sar-NCA (10.0 mg, 0.087 mmol) was dissolved in DCM (178.3 μL) containing benzonitrile (21.7 μL) as the internal standard, into which the DCM solution of Hex-NH_2 (0.1 M, 17.4 μL) was added to start the polymerization. Final condition: $[M]_0 = 0.4$ M, $[M]_0/[I]_0 = 50$. The polymerization solution was transferred into a KBr liquid transmission cell (pathlength = 0.1 mm), which was then subjected to *in situ* FTIR analysis. The conversion was calculated based on the standard curve of Sar-NCA at 1853 cm^{-1} by comparing with the internal standard (i.e., cyano group at 2231 cm^{-1}).

The polymerization kinetics of other monomers (i.e., *N*-Bz-NCA), initiated by other initiators (i.e., PSar macroinitiators), and in other solvents (i.e., DMF) was carried out in a similar way.

2.6. Analysis of PISA process

The PISA kinetics was monitored *in situ* in CD_2Cl_2 by ^{1}H NMR. PSar macroinitiators were prepared first in CD_2Cl_2, which were then used for the second block chain extension. Typically, Sar-NCA (24.2 mg, 0.210 mmol) was dissolved in CD_2Cl_2 (308.2 μL), into which the CD_2Cl_2 solution of isoamylamine (0.1 M, 41.9 μL) was added. At the end of polymerization of the first block, the CD_2Cl_2 solution (357.4 μL) containing *N*-Bz-NCA (20 mg, 0.10 mmol) was added. The mixture was then vortexed and transferred into an NMR tube, and the ^{1}H NMR spectra were monitored at different time intervals. To determine the conversion of NNCA, the integral of α-H signal of *N*-Bz-NCA (δ = 3.97 ppm) was normalized compared to the α-H signal at $t = 0$ (i.e., 100 % remaining NNCA).

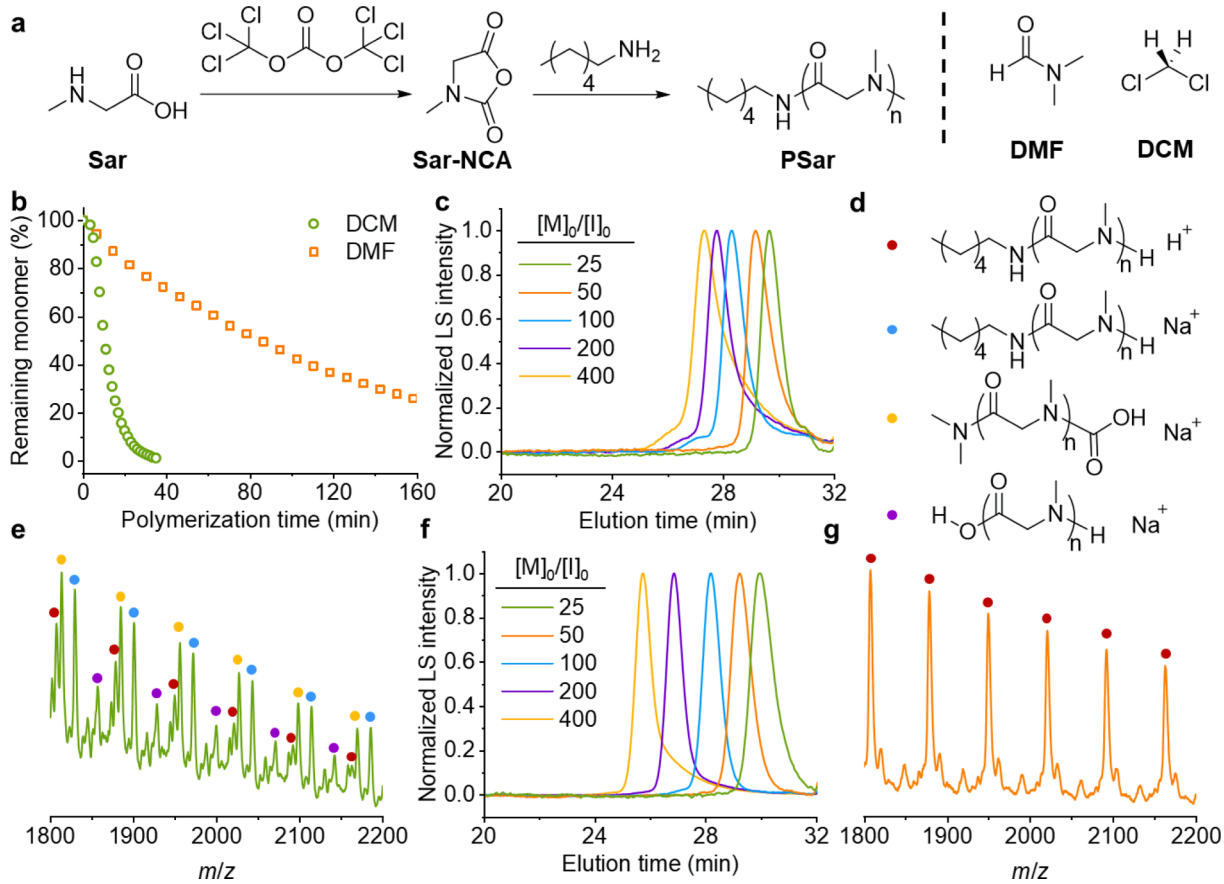

Fig. 1. Solvent-dependent polymerization of Sar-NCA. (a) Synthetic routes to PSar. (b) Conversion of Sar-NCA in anhydrous DCM and DMF under open-air conditions. $[M]_0 = 0.3$ M, $[M]_0/[I]_0 = 50$. (c, f) Normalized GPC-LS traces of PSar obtained in DMF (c) and DCM (f) at various $[M]_0/[I]_0$ under open-air conditions. $[M]_0 = 0.6$ M. (d) Chemical structures of possible polymeric species during MALDI-TOF MS analysis. (e, g) MALDI-TOF spectrum of PSar obtained in DMF (e) and DCM (g) under open-air conditions.

Additionally, the chain length of the second poly(*N*-benzyl-glycine) (PNBz) block was also quantified by ^{1}H NMR. At different time intervals during PISA process, an aliquot of the polymerization mixture was taken out, purified by precipitation in ether/*n*-hexane (1:1, v/v) to remove unreacted monomers, and dried. The resulting polymeric residues were re-dissolved in trifluoroacetic acid-*d* (TFA-*d*) for the quantification of chain length. The degree of polymerization (DP) of PNBz was obtained by comparing the integral ratios between phenyl protons ($\delta \sim 7.24$ ppm) from PNBz block and methyl peak ($\delta \sim 3.16$ ppm) from PSar block.

DLS was used to quantify the changes of nano-assemblies during the PISA process. At different time intervals, an aliquot of the DCM solution/suspension (8 μL) was taken out from the PISA mixture, diluted with DMF (1.0 mL), and subjected to DLS analysis. The size, polydispersity index (PDI), and derived count rate (DCR) information were recorded throughout the PISA process.

2.7. Purification of PISA nano-assemblies

After the PISA process, the solution was diluted with DMF and DCM was removed under vacuum. The solution was then changed to aqueous environment through dialysis. The resulting dispersion of copolypeptoid nano-assemblies was stored in a 4 °C refrigerator. The nano-assemblies in DCM (before dilution with DMF), DMF (after removal of DCM, before dialysis), and water (after dialysis) were subjected to DLS and TEM tests.

2.8. Cytotoxicity studies

RAW 264.7 cells were seeded in a 96-well plate at 1×10^4 cells/well and cultured at 37 °C overnight. $PSar_{50}$-$PNBz_{25}$ nano-assemblies were then added so that the final concentration was 1 μg/mL ~ 100 μg/mL. The cells were then incubated with nano-assemblies in DMEM at 37 °C for 4 h. The cell viability was determined through the MTT assay (λ = 490 nm). Results were expressed as the percentage viability of control cells that were not treated by nano-assemblies.

2.9. Enzymic stability test of nano-assemblies

The dimethyl sulfoxide (DMSO) solution of Nile Red (1 mg/mL, 8 μL) was added into the aqueous dispersion of copolypeptoid nano-assemblies (4 mg/mL), which was stirred overnight. After the removal of DMSO by dialysis, the nano-assemblies were diluted with phosphate buffer saline (PBS) buffer, into which the solution of trypsin or proteinase K (0.8 mg/mL) was added into the solution. The final concentration of nano-assemblies was 0.1 mg/mL. The concentration of trypsin and proteinase K was 0.4 mg/mL and 0.05 mg/mL, respectively. The mixture was incubated at 37 °C. The fluorescence intensity and DLS were then monitored at different time intervals. Additionally, PSar and proteinase K were dissolved in D_2O, which were incubated at 37 °C and characterized at different time intervals to check the enzymatic degradation.

2.10. Kinetic modelling

Kinetic modelling was conducted following the literature procedures [34–36]. Based on the kinetic model:

$$I + M \xrightarrow{k_i} M^* \quad (1)$$

Table 1
Polymerization of Sar-NCA in DCM and DMF at various $[M]_0/[I]_0$.[a]

Entry	Solvent	$[M]_0/[I]_0$	t (min)[b]	$M_{n,\ GPC}$ (kDa)[c]	$M_{n,\ theo.}$ (kDa)	$Đ$[c]
1	DMF	25	120	2.62	1.82	1.05
2	DMF	50	150	3.82	3.65	1.32
3	DMF	100	240	7.26	7.20	1.05
4	DMF	200	300	9.69	14.3	1.09
5	DMF	400	460	10.8	28.5	1.06
6	DCM	25	10	2.54	1.82	1.27
7	DCM	50	15	3.48	3.65	1.12
8	DCM	100	60	7.10	7.20	1.17
9	DCM	200	150	13.5	14.3	1.05
10	DCM	400	190	30.9	28.5	1.08

[a] All polymerizations were conducted in anhydrous solvents at room temperature initiated by Hex-NH_2 under open-air conditions. $[M]_0 = 0.6$ M.
[b] Polymerization time reaching > 95 % monomer conversion.
[c] Determined by GPC with a MALS detector, $dn/dc = 0.0795$.

$$M_i^* + M \xrightarrow{k} M_{i+1}^* \quad i \geq 1 \tag{2}$$

It is straightforward to write down the kinetic equations, with [X] representing the concentrations of various species:

$$\frac{d[I]}{dt} = -k_i[I][M] \tag{3}$$

$$\frac{d[M^*]}{dt} = k_i[I][M] - k[M^*][M] \tag{4}$$

$$\frac{d[M]}{dt} = -k_i[I][M] - k[M]([I]_0 - [I]) \tag{5}$$

$$\frac{d[M_i^*]}{dt} = k[M]([M_{i-1}^*] - [M_i^*]) \quad i \geq 1 \tag{6}$$

where $[I]_0$ represents the initiator concentration at $t = 0$.

3. Results and discussion

3.1. Solvent-dependent polymerization kinetics and MW control of Sar-NCA under open-air conditions

Previous reports emphasized the use of purified DMF for controlled synthesis of PSar [37–39]. In order to check the tolerance of impurities during DMF polymerization, we used commercially available anhydrous DMF as the solvent for the ROP of Sar-NCA under open-air conditions. The polymerization of Sar-NCA was initiated by Hex-NH_2 ($[M]_0 = 0.3$ M, $[M]_0/[I]_0 = 50$) (Fig. 1a), which was monitored by FTIR spectroscopy. In the presence of benzonitrile as an internal standard (Fig. S1), the decrease in anhydride signals at 1853 cm^{-1} and 1788 cm^{-1} suggested the consumption of monomers, reaching ~ 80 % conversion after 160 min (Fig. 1b). The resulting solution appeared yellow that indicated the occurrence of side reactions (Fig. S2). The appearance of the yellow color was attributed to the oxidation of dimethylamine (DMA), which originated from the decomposition of DMF [40]. GPC results of the PSar revealed a monomodal peak with a low MW tailing, likely due to the existence of other undesired polymerization processes that were not initiated by Hex-NH_2 (Fig. 1c). The low-MW tailing on the GPC traces became more pronounced as the $[M]_0/[I]_0$ increased (Fig. 1c), which was attributed to more significant impurity-induced initiations during the elongated polymerization time. While the obtained MWs of PSar 25-mer, 50-mer, and 100-mer were close to their theoretical values, the obtained MWs were only 68 % and 38 % of the theoretical MWs at $[M]_0/[I]_0 = 200$ and 400, respectively (Fig. 1c and Table 1). MALDI-TOF MS analysis further confirmed the undesired initiations, where multiple peaks were observed that corresponded to the initiation by DMA and water (Fig. 1d, 1e, and Fig. S3). While anhydrous DMF was used, the introduction of moisture was inevitable during the dissolving and mixing of monomers and initiators under open-air conditions, leading to monomer hydrolysis and subsequent polymerization [41]. Additionally, the polymerization in DMF conducted in a glovebox, free of oxygen and water, led to a colourless rather than yellow solution (Fig. S2). Nevertheless, MALDI-TOF MS results still indicated the presence of polymeric species from DMA initiation (Fig. S3). Therefore, extensive removal of nucleophilic impurities was necessary to obtain high-quality PSar in a controlled manner [37,39].

We have previously reported the cooperative covalent polymerization (CCP) of amino acid *N*-carboxyanhydrides (NCAs) in solvents with low polarity and weak hydrogen bonding ability, such as chloroform and DCM, which proceeded at an accelerated rate that outpaced various side reactions, resulting in well-defined polypeptide materials [42]. Inspired by the CCP results, we reasoned that the polymerization of Sar-NCA would show similar solvent effect due to the polar monomers and polymer chains. The weaker solvation of propagating chains and monomers in chlorinated solvents enhanced their interactions, leading to a faster polymerization rate that inhibited undesired initiations. Moreover, the immiscibility between water and CCP solvents also facilitated the removal of water from the polymerization mixture. Therefore, we studied the polymerization of Sar-NCA in DCM, a common CCP solvent for NCA polymerization, under open-air conditions. As shown in Fig. 1b, the polymerization of Sar-NCA in DCM proceeded in a much faster manner than that in DMF under identical conditions, reaching > 98 % conversion after 30 min at $[M]_0 = 0.3$ M (Fig. 1b). The solvent-dependent polymerization behaviours agreed well with the recent reports by Lu and co-workers [25]. The fast polymerization

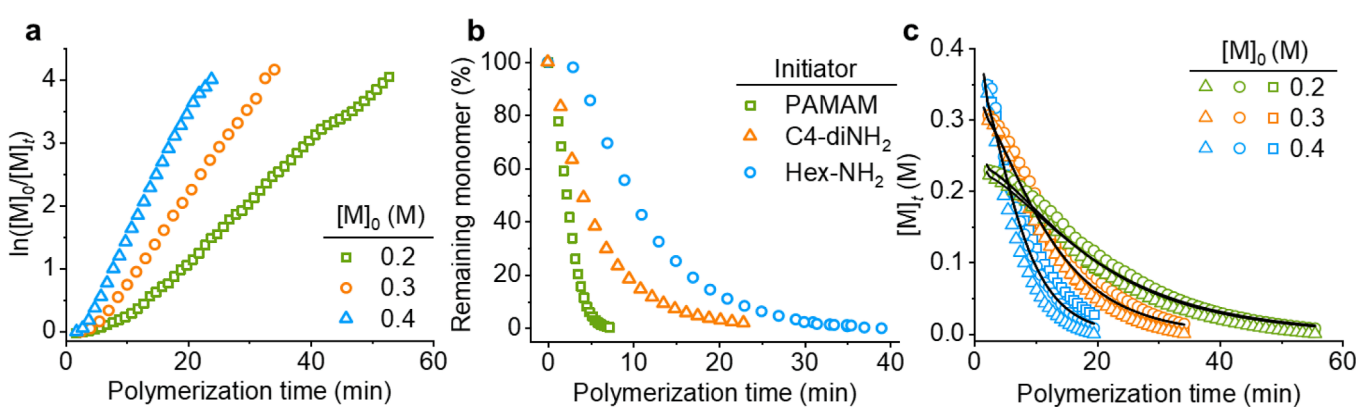

Fig. 2. Polymerization behaviour of Sar-NCA in DCM. (a) Semilogarithmic kinetic plot of polymerization of Sar-NCA in DCM at various $[M]_0$. $[M]_0/[I]_0 = 50$. (b) Conversion of Sar-NCA initiated by various initiators. $[M]_0 = 0.2$ M, $[M]_0/[I]_0 = 50$. $[I]_0$ is the concentration of initiating $-NH_2$. (c) Fitting (black lines) of the $[M]_0$-dependent polymerization kinetics of Sar-NCA with the kinetic model. $[M]_0/[I]_0 = 50$.

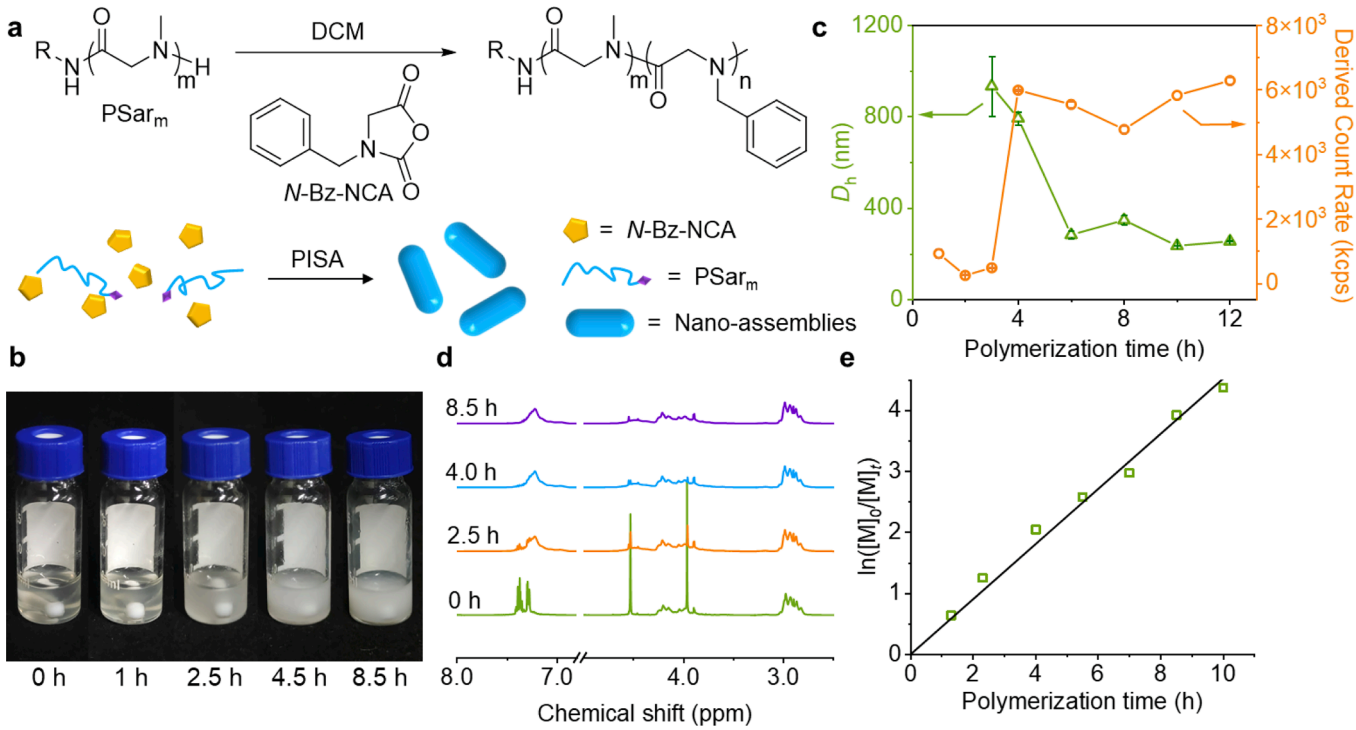

Fig. 3. Preparation of copolypeptoid nano-assemblies via PISA in DCM. (a) Scheme showing the PISA process with PSar macroinitiators and *N*-Bz-NCA monomers. (b) Photographs illustrating the turbidity change during PISA. (c) Change in D_h and DCR during the PISA process ($n = 3$). (d) NMR spectra showing the consumption of *N*-Bz-NCA monomers over time. (e) Semilogarithmic kinetic plot of polymerization of *N*-Bz-NCA monomer during PISA. $[M]_0/[I]_0 = 25$, solid content = 15 %.

kinetics helped minimize undesired initiations, as the GPC characterization of resulting PSar revealed monomodal peaks with low dispersity ($Đ = M_w/M_n < 1.2$) at different $[M]_0/[I]_0$ ratios (Fig. 1f), indicating well-defined polymerization progress. In sharp contrast to the PSar synthesized in DMF, the low-MW tailing was observed only at $[M]_0/[I]_0 = 400$ in DCM. As a result, the obtained MWs agreed well with the theoretical values in the range of $[M]_0/[I]_0 = 25$–400 (Fig. 1f and Table 1). MALDI-TOF analysis further confirmed insignificant initiations from impurities, as only Hex-NH_2-initiated PSar chains were observed (Fig. 1d, 1 g).

3.2. Polymerization behaviour of Sar-NCA in DCM

The improved MW control of PSar from accelerated polymerization encouraged us to further explore the polymerization behaviour of Sar-NCA in DCM, aiming to draw an analogy with CCP of NCA. The polymerization of Sar-NCA in DCM became faster as the $[M]_0$ increased (Fig. 2a), suggesting the existence of cooperative interactions between propagating chains or between propagating chains and monomers [35,36]. The $[M]_0$-dependent polymerization behaviour enabled the preparation of polypeptoid materials at higher $[M]_0$ in a shorter time. Well-defined PSar 50-mer was obtained within 15 min at $[M]_0 = 0.6$ M (Fig. S4 and Table S1). Additionally, the enhanced cooperative interactions would also enable accelerated kinetics from initiators with multiple initiating sites [42]. For instance, 1,4-butylenediamine (C4-diNH_2) and PAMAM (Generation 2), which had 2 and 16 initiating sites in one molecule, respectively, were used as the initiators for the polymerization of Sar-NCA in DCM. As expected, the polymerization rate increased as the density of amino groups increased (Fig. 2b, Fig. S4, and Table S2), indicating similar cooperative interactions with CCP. While the backbones of polypeptoids and polypeptides were different, both showed relatively high polarity with the amide structures. The use of DCM with weaker solvation of polymer chains thus enhanced the chain-chain interactions or chain-monomer interactions, leading to the accelerated rate when the initiating sites were in close proximity [42,43].

Despite the similar polar polymer backbones, polypeptoids differed from polypeptides that polypeptoids lack the hydrogen-bonding donating N–H groups on their backbones. Therefore, the hydrogen-bonding-mediated monomer-polymer interactions observed in CCP would not contribute to the rate acceleration [35]. Indeed, the addition of 18-crown-6, a crown ether (CE) catalyst that significantly accelerated the NCA polymerization [44], did not show any catalytic effect in the polymerization of Sar-NCA (Fig. S4). The *N*-Me group at the N terminus likely blocked its close interactions with CE and Sar-NCA monomer, leading to unsubstantial rate acceleration.

3.3. Kinetic modelling

Due to the absence of a chiral centre, PSar was unable to form ordered secondary structures like polypeptides [45,46]. Therefore, the unique two-stage polymerization kinetics of CCP induced by coil-to-helix transition was not observed during the polymerization of Sar-NCA. Indeed, CD characterization revealed a flat curve of PSar in DCM (Fig. S5), confirming the lack of ordered conformation. The kinetic curves initiated from Hex-NH_2 and PSar macroinitiators were also comparable (Fig. S5), substantiating the one-stage kinetics. The simple kinetic model was therefore used to fit the polymerization of Sar-NCA, which is described by the following reactions:

$$I + M \xrightarrow{k_i} M^* \quad (1)$$

$$M_i^* + M \xrightarrow{k} M_{i+1}^* \quad i \geq 1 \quad (2)$$

where M stands for Sar-NCA monomer and M_i* represents a polypeptoid chain with DP of i and an active site (*) at the chain-end. The initiation and propagating stages proceed with a rate constant of k_i and k, respectively. By sharing the same k_i but allowing k to be individually optimized, the $[M]_0$- and $[I]_0$-dependent kinetic curves were fit with the kinetic model (Fig. 2c and Fig. S6). The propagating rate constant was

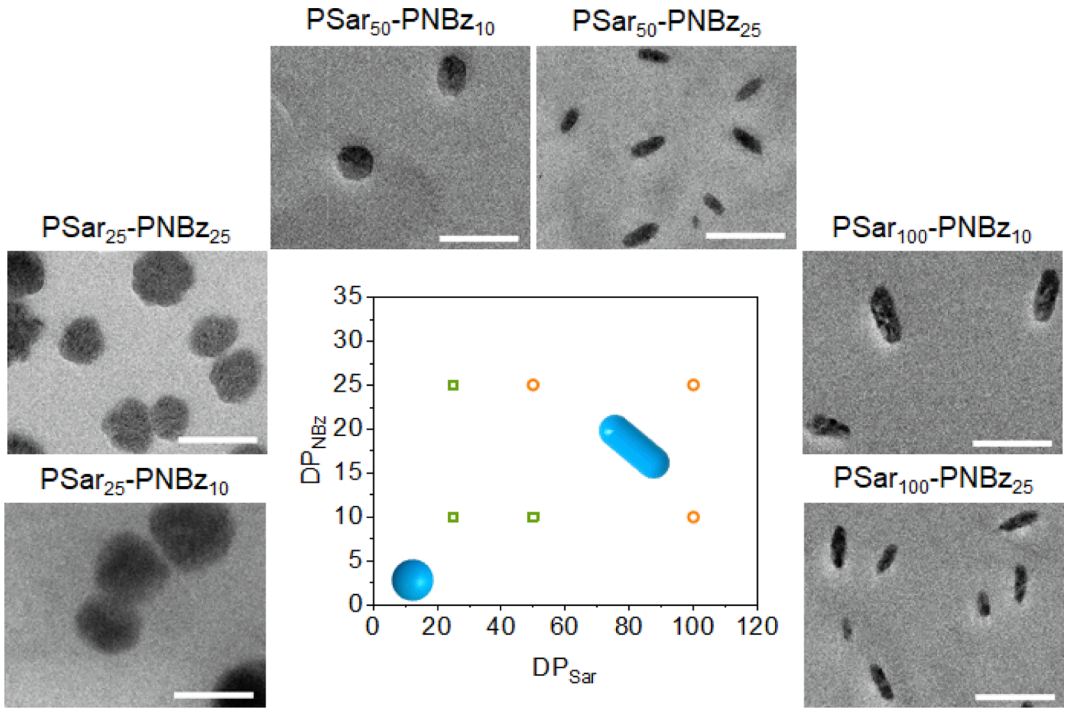

Fig. 4. Phase diagram of the PSar-PNBz nano-assemblies prepared by PISA in DCM. Solid content = 15 %. Scale bar = 500 nm. Green squares represent spherical assemblies and orange circles represent non-spherical assemblies. TEM images were incorporated to show the block-length-dependent morphology of PSar-PNBz nano-assemblies.

calculated to be (0.23 ± 0.01), (0.31 ± 0.01), and (0.57 ± 0.01) M^{-1} s^{-1} for the polymerization at $[M]_0$ = 0.2, 0.3, and 0.4 M, respectively (k_i = 0.01 M^{-1} s^{-1}), substantiating the $[M]_0$-dependence in DCM as a result of possible interchain interactions, aggregations or enhanced chain-monomer interactions.

3.4. PISA of NNCA for the preparation of nano-assemblies

The solvent-dependent polymerization behaviour was also evaluated with another NNCA monomer, *N*-Bz-NCA (Fig. S7), which was selected due to the ease of purification by recrystallization [47]. The *in situ* FTIR analysis suggested a slower polymerization rate than that of Sar-NCA, resulting from the larger steric hindrance from the side-chain benzyl groups [48,49]. Meanwhile, similar solvent-dependent kinetics was observed for *N*-Bz-NCA, where the consumption of half monomer was achieved after 75 and 600 min for the polymerization in DCM and DMF, respectively (Fig. S7). Moreover, precipitation of the resulting polypeptoids, PNBz, was observed during the polymerization in both solvents (Fig. S7).

Inspired by the solubility difference of *N*-Bz-NCA monomer and PNBz polymer, we developed a PISA system to prepare block copolypeptoid nanoparticles in one pot (Fig. 3a). Previous reports on this copolymer resulted in mainly aggregates due to the strong interactions between PNBz segments [47]. PSar 50-mer was first prepared in DCM (Fig. S8), which was then used as the macroinitiator and the stabilizer for the subsequent polymerization of *N*-Bz-NCA ($[M]_0/[I]_0$ = 25, solid content = 15 %). As the polymerization proceeded, the reaction mixture gradually turned cloudy after 2.5 h (Fig. 3b), and became more turbid as the polymerization time increased, suggesting the formation of block copolypeptoid nanoparticles. In order to monitor the nanoparticle formation, an aliquot of the mixture was taken out at different time intervals and diluted with DMF, and the change in DCR, which was positively correlated to the nanoparticle concentration [27,50,51], was monitored through DLS analysis. DCR remained low in the first 3 h of polymerization that suggested few nano-assemblies, but significantly increased at 4 h post initiation (Fig. 3c), indicating the formation of copolypeptoid nanoparticles that was consistent with the turbidity change. The initial size of the nanoparticles was extremely large (~ 1 μm) with a large PDI ~ 0.8, showing fluctuating signals between tests. Stable signals were obtained 6 h post initiation, which showed a size of ~ 280 nm with a PDI ~ 0.2.

The PISA kinetics was monitored *in situ* with NMR, which exhibited an obvious decrease in signals at 3.97 and 4.53 ppm (Fig. 3d), corresponding to the benzylic protons and α-protons of *N*-Bz-NCA, respectively. Quantitative analysis revealed a smooth kinetic curve with no obvious turning point (Fig. 3e). This result resembled the PISA of NCA [27,51], indicating good solubility of *N*-Bz-NCA in DCM. While the monomer consumption reached ~ 50 % after 1 h, no significant turbidity was observed in the first 2 h, suggesting a relatively slow assembly kinetics. Additionally, the smooth polymerization kinetics was also confirmed by *in situ* FTIR kinetics, before significant turbidity occurred (Fig. S9). The PISA proceeded with a much faster kinetics in DCM than that in DMF, which was consistent with the solvent-dependent polymerization behaviour of *N*-Bz-NCA. As a result, later PISA studies were only conducted in DCM. Meanwhile, the PISA progress was quenched at different time intervals through the addition of trifluoroacetic acid, and the block copolypeptoids PSar-PNBz were purified and analysed by NMR (Fig. S9). The MW of the second PNBz block exhibited a linear relationship with the conversion of *N*-Bz-NCA, suggesting a living PISA process (Fig. S9).

It has been reported that both the block ratio and the solid content played an important role in determining the morphology of PISA nano-assemblies [26]. Therefore, the PSar-initiated PISA of *N*-Bz-NCA was first conducted with different PSar lengths and $[M]_0/[I]_0$ (solid content = 15 %) (Fig. S8, Table S3, and Table S4). After the monomer conversion reaching > 99 %, the resulting dispersion was switched to the aqueous phase for later enzymatic and cellular studies. Briefly, the dispersion was diluted with DMF, evaporated under vacuum to remove DCM, and dialyzed against water. The final nano-assemblies in the aqueous phase was then characterized by TEM. With a short length of PSar (i.e., DP = 25), solid nano-assemblies were observed with size ~ 400 nm (Fig. 4), which was much larger than the theoretical diameter of conventional

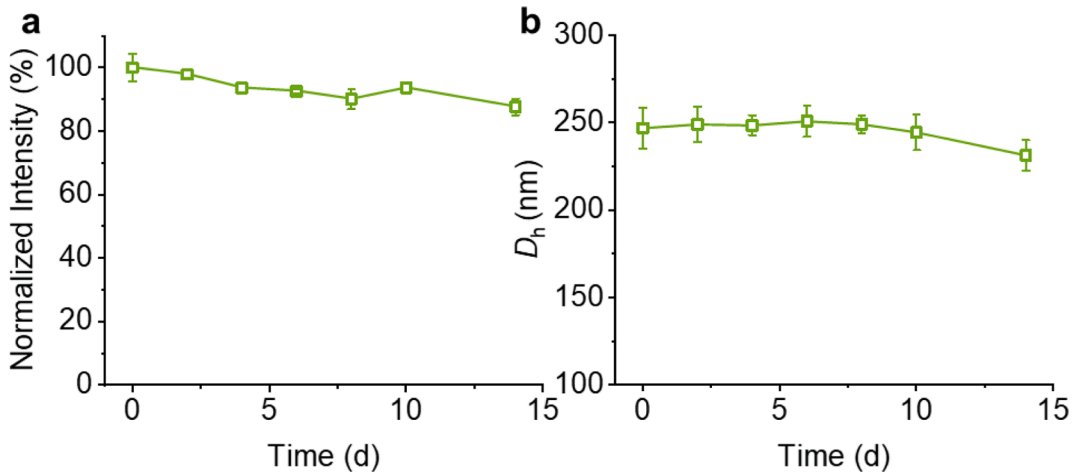

Fig. 5. Stability of $PSar_{50}$ -$PNBz_{25}$ nano-assemblies in PBS at 37 °C against trypsin. Changes in the fluorescent stability of encapsulated Nile Red (a) and nano-assemblies size (b) were monitored after different incubation times (n = 3). λ_{ex} = 540 nm, λ_{em} = 620 nm. c(nano-assemblies) = 0.1 mg/mL, c(trypsin) = 0.4 mg/mL.

spherical micelles from PSar-PNBz. Due to the shorter hydrophilic PSar segments, the block copolypeptoids likely formed inverted micelles with PSar cores and PNBz coronas, which further assembled into compound micelles [52,53]. The increase in PSar length led to smaller assemblies due to the stabilization effect from the PSar block. Specifically, while $PSar_{50}$-$PNBz_{10}$ exhibited a spherical morphology, $PSar_{50}$-$PNBz_{25}$, $PSar_{100}$-$PNBz_{10}$, and $PSar_{100}$-$PNBz_{25}$ showed non-spherical, elongated structures with an aspect ratio ~ 3–5 (Fig. 4 and Fig. S10). Interestingly, $PSar_{50}$-$PNBz_{25}$ exhibited spherical morphology in DCM and DMF before the solvent switch process, even with a similar size (~290–300 nm) as validated by DLS (Fig. S11). While the non-conventional assembly behaviour and the solvent-induced morphology transition of PSar-PNBz required further investigation in our future studies, similar morphology was observed for polypeptide-based assemblies reported by several research groups [15,54,55]. Therefore, the strong interactions between PNBz polypeptoid chains likely contributed to the non-spherical assembly, although no ordered conformation was found from CD analysis (Fig. S12) [56]. Moreover, the rod-like morphology was also confirmed by SEM (Fig. S13). The non-spherical morphology may facilitate the extravasation of PSar-nanocarriers [57], allowing the cargos to reach targeted tissues in a faster manner. Moreover, the solvent switch process was also applied to a DCM-soluble diblock copolymer, PSar-*block*-poly (γ-benzyl-$_L$-glutamate), yielding precipitations rather than well-defined assemblies (Fig. S14). Due to serious chain entanglements at such high solid contents (i.e., 15 %), ordered assembly is difficult, ruling out the possibility that PSar-PNBz assemblies were driven by solvent switch. Meanwhile, the change in solid content also altered the assembly morphology. $PSar_{50}$-$PNBz_{10}$ at a low solid content of 5 %, for instance, assembled into much smaller nanoparticles (Fig. S15).

3.5. Enzymatic degradation studies

With the backbone *N*-substitutions, polypeptoids were reported to be resistant to protease degradations, which was beneficial for drug carrier designs, especially when combined with the stealth effect of PSar. PSar-PNBz exhibited low cytotoxicity, with > 85 % cell viability observed in the presence of up to 100 μg/mL nano-assemblies (Fig. S16). In order to check the enzymatic stability of PSar nano-assemblies, Nile Red was encapsulated into copolypeptoid nanoparticles as a fluorescence probe, whose fluorescence intensity was highly dependent on the hydrophobicity of the environment [58]. Upon the treatment with trypsin, the fluorescent signals of Nile Red only exhibited 10 % decrease after 14 days of incubation at 37 °C (Fig. 5a), indicating that most copolypeptoid nanoparticles remained stable. On the other hand, the fluorescent signals of Nile Red exhibited a 20 % decrease in the presence of proteinase K (Fig. S17), a broad-spectrum protease that hydrolysed a variety of peptide bonds. The decrease in Nile Red signals was presumably attributed to the interference of assembly behaviours of copolypeptoids, since further NMR studies suggested negligible degradation of the polypeptoid chains (Fig. S18). Additionally, DLS analysis agreed well with the Nile Red experiment, where the size of PSar-PNBz assemblies remained unchanged for 14 days (Fig. 5b), highlighting the use of the nanoparticles for drug delivery applications with minimized premature cargo release.

4. Conclusions

In summary, we reported the preparation of copolypeptoid nano-assemblies bearing PSar corona through PISA in DCM. The decreased solvation facilitated the chain-chain or chain-monomer cooperative interactions, leading to accelerated polymerization kinetics in chlorinated solvents. Taking advantage of the solubility change of *N*-Bz-NCA after polymerization, nano-assemblies with different morphologies and solid contents were prepared within several hours. With the facile preparation of PSar-decorated nano-assemblies at high solid contents, one can easily access drug-loaded nanomedicine with potential long-circulation properties as alternatives of PEGylated nanoparticles, benefiting the downstream studies on the biological fate and controlled cargo release.

CRediT authorship contribution statement

Qing Gao: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Hailin Fu:** Methodology, Investigation, Formal analysis. **Hui Liu:** Investigation, Formal analysis. **Pengfei Li:** Investigation, Formal analysis. **Jin Shi:** Investigation, Formal analysis. **Yao Lin:** Writing – review & editing, Methodology, Formal analysis. **Ziyuan Song:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113720.

Data availability

Data will be made available on request.

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